Neural Heterogeneity of Word Response

Proposal?

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Gap

Studies of free recall describe the average response to a word, with high gamma rising first in visual regions and later in prefrontal regions (Burke et al., 2014; Kucewicz et al., 2019), and theta, alpha/beta and gamma power largely rising at separate sites, in separate sequences (Marks et al., 2021). These studies average over word events and rarely account for the heterogeneity across electrodes in a region, or other factors contributing towards variability in responses. How much of the response differs between subjects, electrodes, sessions and word events, which part is a stable property of a site, and what changes it from one word event to the next have not been measured. With 361 subjects, 1007 sessions and ∼200 words per session, this dataset can answer these questions. The contribution of task structure towards the memory-related component has been identified by DeHaan and Halpern (TBD), which the main mixed-effects model used here draws from.

Aims

Aim 1. How much of the response is stable, and how much changes from one word to the next? On average, occipital high gamma rises at word onset ($d = 0.34$) and alpha falls (-0.10), with smaller changes elsewhere (Fig. 1A). Split the response to each word event (word on vs. word off, Cohen's d effect size) into subject, electrode, session, task and word-event components per region, with a random-intercept model. *Preliminary (up to 5 electrodes per subject and ROI; Fig. 1B, C):* most variance across word events is unique to an electrode during a word event (alpha 71–86%, high gamma 56–89%, latency 82–94%). The next largest part is a word event's deviation shared by all of an ROI's electrodes (alpha 11–25%, high gamma 6–15%, latency 4–13%). Session, list in session and task structure are each small (≤3%). Stable differences are largest in occipital high gamma (subject 8%, electrode 22%, electrode in session 3%); for alpha outside occipital regions subject and electrode together are ≤2%, and for latency ≤6%. Adding the anatomical subregion within a lobe (Fig. 1B, C) takes about 3% of occipital high-gamma variance, out of the electrode and subject components, none of alpha's, and ≤2% of latency's.

Planned: the same split for power in the pre-stimulus period alone, as a control.

Aim 2. Which electrodes respond in which band, and do they covary? Outside occipital regions 68–76% of electrodes show neither an alpha nor a high-gamma response, against 35% in occipital regions (paired t over word events, FDR 0.05); 31% of occipital electrodes respond in both bands, 5–11% elsewhere. Which electrodes respond strongly within a band replicates

between session halves (Fig. 2A; median $r = 0.90$ in occipital high gamma, 0.80 in occipital alpha, 0.20–0.59 elsewhere). Across electrodes the bands share the size of the response more than its direction (Fig. 2B): r of $|d|$ is 0.46 in occipital regions and 0.13–0.21 elsewhere, while r of signed d is between -0.11 and 0.15. In occipital regions, where high gamma rises, alpha falls at 19% of electrodes and rises at 8%. Marks et al. (2021) found the bands at separate sites; how they covary depends on the timescale (Fig. 2C). Along the list high gamma falls in every ROI (-0.004 to -0.012 d per position) and alpha rises in every ROI but the hippocampus (0.004–0.005 d per position), and in frontal, parietal and occipital regions the two serial-position curves are anticorrelated ($r = -0.12$ to -0.20), while from one word event to the next the bands are nearly independent (-0.01 to 0.04). In limbic regions and the hippocampus they are weakly positively related at every timescale (0.07–0.13).

Planned: does an electrode's latency replicate, also the anatomy of these response types within a region.

Aim 3. What links one word to the next? After a long pre-stimulus period the occipital high-gamma peak is larger (Fig. 3A; +0.058, FDR 0.05), while occipital high gamma before the word is slightly lower (-0.010): the effect is in the response, not a pre-stimulus level carried into it. At the same electrode, with serial-position means removed, the previous word's response correlates with the current one in every ROI (Fig. 3B; high gamma $r = 0.06$–0.12, alpha 0.03–0.05), more than the response two words back does (by 0.014–0.024 in high gamma and 0.007–0.020 in alpha; paired t, FDR 0.05): a slow drift plus a carry-over from the previous word. The carry-over does not change with the length of the pre-stimulus period (long − short $|\Delta r| \leq 0.008$, n.s.), so it does not explain the effect of a long pre-stimulus period.

Planned: what the shared word-event component is (arousal, attention, or a common reference), whether the effect of a long pre-stimulus period is recovery from adaptation (Groen et al., 2022) or temporal expectation, and latency per word event.

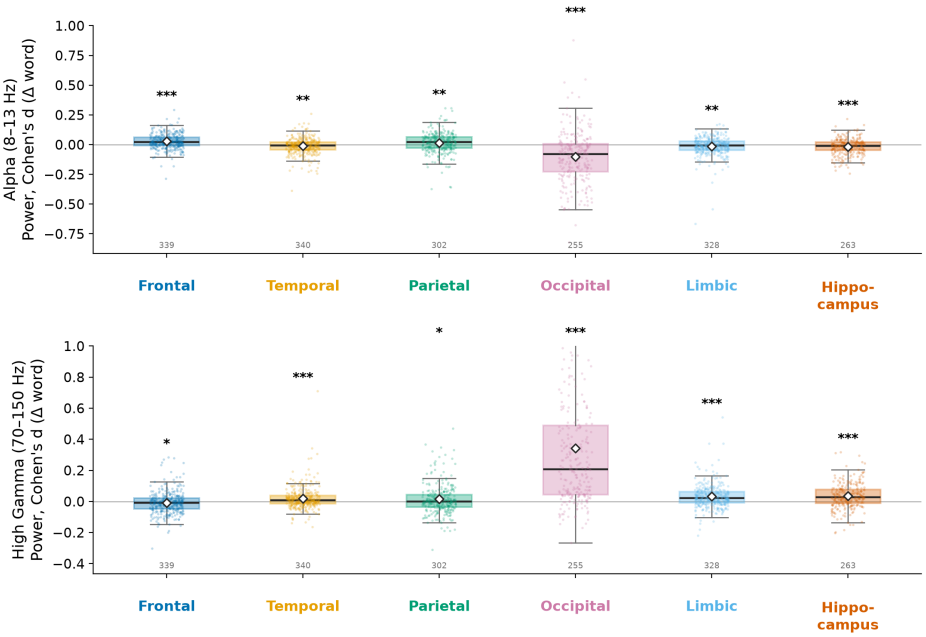
Aim 4. Does connectivity between regions vary in the same way? At word onset alpha ciPLV rises in 9 of 12 regions, with hubs first in occipital regions (0–400 ms) and then in limbic, parietal, frontal and temporal regions (Fig. 4A).

Planned: the variance split of Aim 1 for synchrony (Fig. 4B).

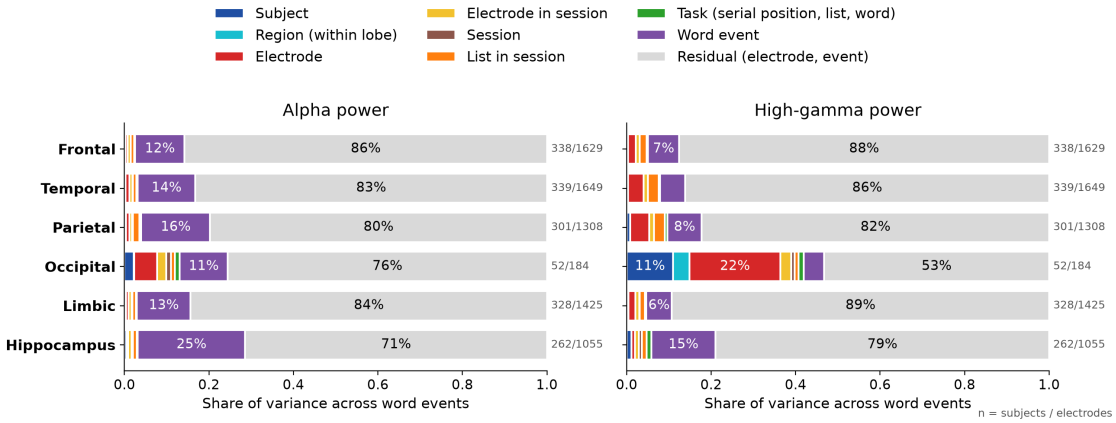
Summary. Differences between locations are stable but large only in occipital high gamma; elsewhere most of the variance is from one word event to the next.

Figures

A (Done)



B (Prelim, 5 electrodes per subject/ROI)



C (Prelim, 5 responsive electrodes per subject/ROI)

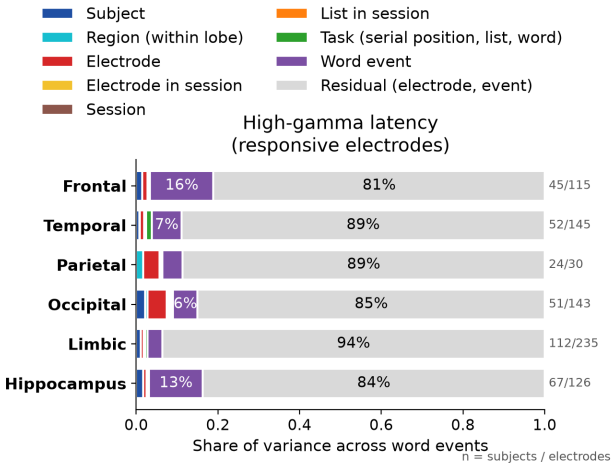
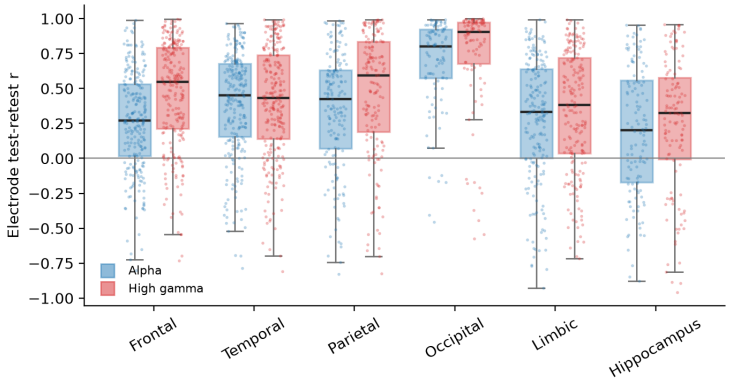
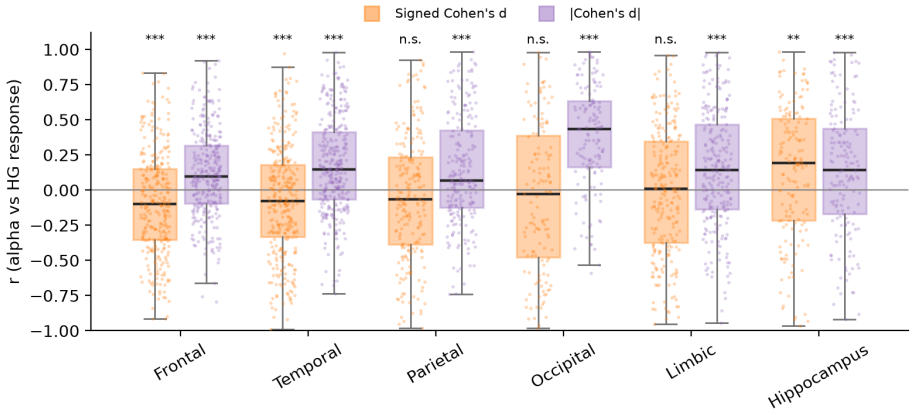


Figure 1: (A) Power change at word onset (Cohen's d, word on vs. off, across subjects). (B) Variance components of the response to each word event, one mixed model per ROI (Aim 1); occipital: with anatomical subregion as a component (52 subjects). (C) As B for high-gamma latency (responsive electrodes); frontal, temporal, parietal and occipital: with anatomical subregion (regions with ≥30 subjects).

A (Done)



B (Done)



C (Done)

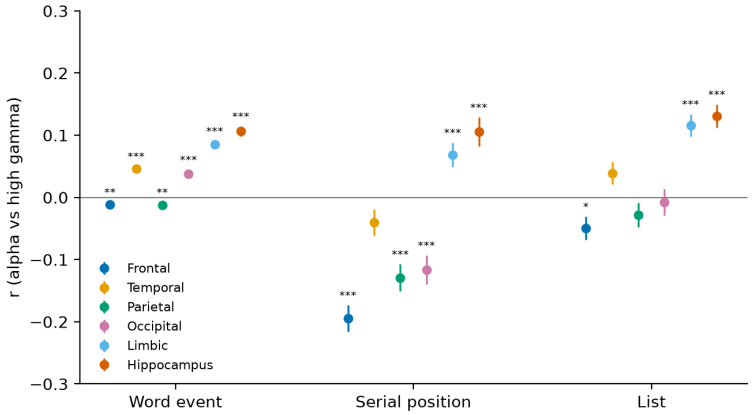
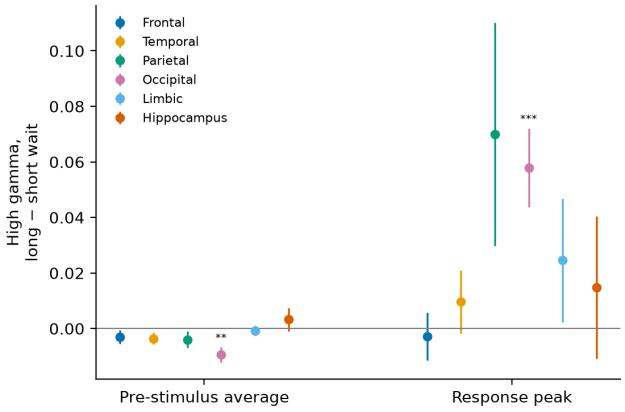


Figure 2: (A) Electrode test-retest: r of electrodes' Cohen's d between session halves (1 dot = 1 subject). (B) Alpha vs. high gamma across an ROI's electrodes: r of Cohen's d and of $|d|$ per subject (dots); stars: subjects' mean r vs. 0, FDR. (C) Alpha vs. high gamma by timescale: across word events at the same electrode (serial-position and list means removed), across serial-position means and across list means; mean r across subjects ± SEM per region, stars: FDR.

A (Done)



B (Done)

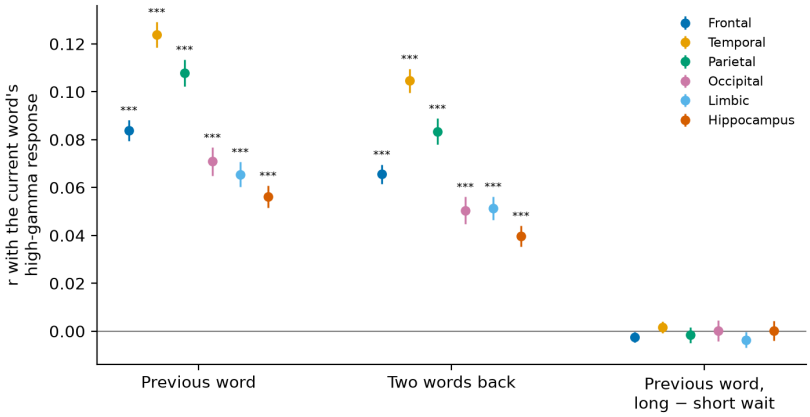
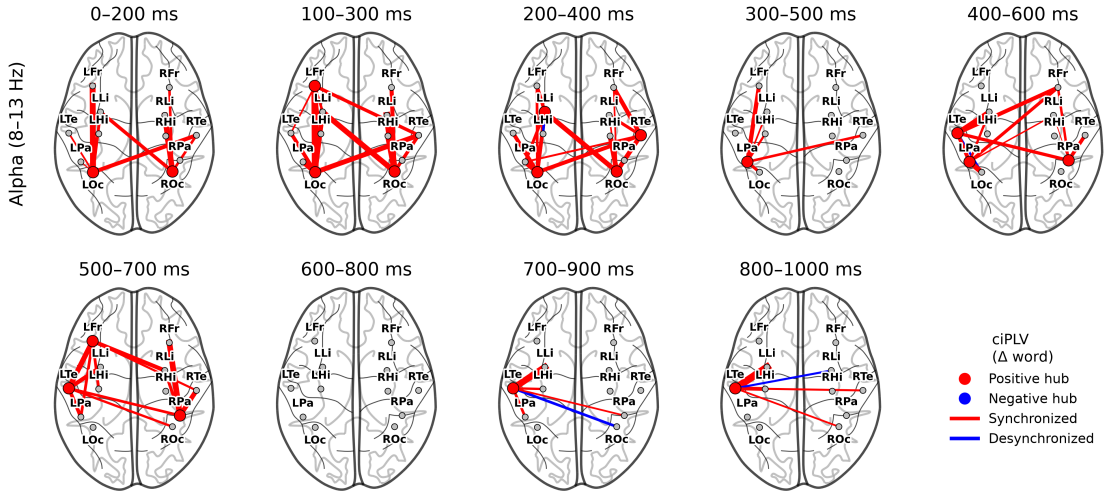


Figure 3: (A) High gamma after long vs. short blanks: pre-stimulus average (−700 to −100 ms) and response peak (responsive electrodes). (B) High-gamma response of each word vs. the word one and two back in the same list (same electrode, serial-position means removed), and the change after long vs. short waits. Mean across subjects ± SEM per region; stars: FDR.

A (Done)



B (Planned)



Figure 4: (A) Alpha ciPLV change at word onset in 200 ms windows: hubs (nodes) and each hub's five strongest connections. (B) Variance components of synchrony.

References

Burke, J. F., Long, N. M., Zaghloul, K. A., Sharan, A. D., Sperling, M. R., and Kahana, M. J. (2014). Human intracranial high-frequency activity maps episodic memory formation in space and time. *NeuroImage*, 85:834–843.

DeHaan, R. D. and Halpern, D. (TBD). *On the specificity, causality, and reliability of electrophysiological correlates of human memory: limitations and opportunities*. PhD thesis, University of Pennsylvania.

Groen, I. I. A., Piantoni, G., Montenegro, S., Flinker, A., Devore, S., Devinsky, O., Doyle, W., Dugan, P., Friedman, D., Ramsey, N. F., Petridou, N., and Winawer, J. (2022). Temporal dynamics of neural responses in human visual cortex. *The Journal of Neuroscience*, 42(40):7562–7580.

Kucewicz, M. T., Saboo, K., Berry, B. M., Kremen, V., Miller, L. R., Khadjevand, F., Inman, C. S., Wanda, P., Sperling, M. R., Gorniak, R., Davis, K. A., Jobst, B. C., Lega, B., Sheth, S. A., Rizzuto, D. S., Iyer, R. K., Kahana, M. J., and Worrell, G. A. (2019). Human verbal memory encoding is hierarchically distributed in a continuous processing stream. *eneuro*, 6(1):ENEURO.0214–18.2018.

Marks, V. S., Saboo, K. V., Topçu, Ç., Lech, M., Thayib, T. P., Nejedly, P., Kremen, V., Worrell, G. A., and Kucewicz, M. T. (2021). Independent dynamics of low, intermediate, and high frequency spectral intracranial eeg activities during human memory formation. *NeuroImage*, 245:118637.